

Day 9 — Speed Sprints + Project Start

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will demonstrate sub-10-second mental Nikhilam multiplication on Day 1's baseline problem (and similar) AND will have a project plan/draft ready for Day 10's showcase.

Materials

- Day 1 baseline sheet (per student) — they pull it from their workbook.
- Sprint sheet: 10 Nikhilam-suitable problems, timer-friendly.
- Project planning slip.
- `activities/activity-02-speed-sprint.md` worksheet.

Lesson flow

Warm-up (5 min)

- Complement-to-100 + excess-from-100 mixed round.
- “*Today is your retest day. Get sharp.*”

Speed sprint (15 min) — `activities/activity-02-speed-sprint.md`

Brief: - 10 Nikhilam problems on a sheet, mixed (both-below, both-above, mixed-sign, near-1000). - 3 minutes individual. Solve as many as you can. - 1 minute self-correction. - Repeat (round 2): swap problems on a fresh sheet, 3 minutes again. - 1 minute self-correction. - 3-minute paired chorus round: partner reads, the other answers in chorus.

Day 1 retest (5 min)

- Students pull their Day 1 baseline sheet ($97 \times 96 = 9312$).
- On a fresh strip, redo the problem using Nikhilam.
- Record new time.
- Compare: **Day 1 (long) vs. Day 9 (Nikhilam).**

Most students will go from 30–60 sec → under 10 sec. Some will hit 3 sec.

Project planning (15 min)

- Distribute project planning slip (see `assessments/project-brief.md`).
- Each pair fills out:
 - Format chosen (video / workbook / data project).
 - One worked example they'll feature.
 - Materials needed.
 - Who does what.

- Teacher circulates, gives quick approval on viability.

Wrap-up (5 min)

- Brief share-out: 3 pairs announce their plan in one sentence.
- Reminder: project must demonstrate the *Nikhilam* method correctly; must cite at least one source PDF.

Student-facing content

Day 1 retest:

My time on 97×96 (Day 1, long multiplication): ____ seconds My time on 97×96 (Day 9, Nikhilam): ____ seconds **My speedup:** ____×

Sprint scoring: 10 problems in 3 minutes. Each correct = 1 point. Each wrong with right method = 0.5 (you knew the method). Each wrong with wrong method choice = 0.

Project plan (fill in): - Format: _____ - Featured example: _____
- Materials: _____ - Partner / role split: _____

Homework

- Finish project plan at home.
- Practise your featured example until you can present it without notes.
- 5 mental Nikhilam problems a day, every day, even after this module ends.

Differentiation

- **Tier up:** Sprint round of 20 problems in 3 minutes (full speed).
- **Tier down:** Sprint sheet with 5 problems in 3 minutes (slower pace, still timed).

Teacher notes

- The retest moment is the most powerful one of the module. Make space for it — even quiet students will smile when they see their improvement.
- Some pairs won't have a clear project plan by end of class. Pull them aside for 5 min after — better to spend the time now than have a shaky project on Day 10.
- Confirm the data-project pairs have at least 5 willing test subjects lined up.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* — sprint problem source.
- See `assessments/project-brief.md` for the full project specification.